

Sourabh Joshi

VNIT Nagpur, India | joshisourabh47@gmail.com | +91 96600 88773
sourabhjoshi.info | linkedin.com/in/sourabhk-nitb | github.com/sourabhjoshi1

Professional Summary

M.Tech Electrical Engineering student at VNIT Nagpur with experience in power electronics, electrical drives, battery management systems, control systems, and industrial electrical maintenance. Skilled in MATLAB/Simulink, Python, machine learning, DC-DC converters, Li-ion battery SoC/SoH estimation, SiC/GaN wide-bandgap devices, and power conversion systems.

Education

Visvesvaraya National Institute of Technology (NIT Nagpur) <i>M.Tech in Electrical Engineering</i>	July 2024 – Expected June 2026 <i>CGPA: 7.22/10</i>
Maulana Azad National Institute of Technology (NIT Bhopal) <i>B.Tech in Electrical Engineering</i>	July 2017 – May 2021 <i>CGPA: 6.91/10</i>

Experience

MSP Steel and Power Limited <i>Graduate Engineering Trainee, Pellet Division – Electrical Department</i>	Jan 2022 – Jan 2023 <i>Raigarh, Chhattisgarh</i>
- Supported operation, preventive maintenance, and troubleshooting of electrical systems in a pellet plant environment. - Worked with motor control centers (MCCs), transformers, automation systems, motors, starters, and plant electrical distribution equipment. - Assisted in identifying electrical faults, coordinating corrective actions, and maintaining reliable operation of production equipment.	

Projects

Deep Learning Architectures for Precise SoC and SoH Estimation in Lithium-Ion Batteries using Bio-Inspired Optimization (M.Tech Thesis) 2026	July 2025 – April 2026
- Developing AI-driven battery state estimation frameworks for Lithium-Ion Battery Management Systems (BMS), focusing on accurate State-of-Charge (SoC) and State-of-Health (SoH) prediction under varying operating and aging conditions. - Implementing and comparing advanced deep learning architectures including LSTM, GRU, and hybrid CNN-LSTM networks for spatiotemporal modeling of battery voltage, current, temperature, and degradation data. - Designing an Enhanced Genetic Algorithm (EGA) for automated hyperparameter optimization of deep learning models, improving estimation accuracy, convergence stability, and robustness across battery life cycles. - Evaluating model performance using RMSE, MAE, and R^2 metrics and benchmarking against conventional estimation methods for intelligent real-time BMS applications in electric vehicles and energy storage systems.	
Evaluation of Wide Bandgap Devices for Electric Vehicle Power Electronics	May 2026 – Jun 2026
- Conducted a research-level evaluation of Silicon (Si), Silicon Carbide (SiC), and Gallium Nitride (GaN) power semiconductor devices for electric vehicle power electronics applications. - Analyzed material figures of merit, conduction and switching loss mechanisms, thermal behavior, packaging constraints, electromagnetic interference (EMI), and reliability challenges associated with wide-bandgap devices. - Investigated the impact of SiC and GaN devices on EV traction inverters, onboard chargers, DC fast chargers, and auxiliary DC-DC converters, focusing on efficiency, power density, thermal management, and system-level performance improvements. - Developed comparative design frameworks and application-oriented analyses highlighting device-to-system co-design considerations for next-generation electric vehicle power conversion systems.	
BLDC Motor Speed Control Using MATLAB/Simulink	Jan 2020 – May 2020
- Developed a closed-loop speed control model for a brushless DC motor using a PID controller in MATLAB/Simulink. - Improved transient response by reducing steady-state error and settling time through controller tuning and simulation analysis.	

Publication

A Comparative Analysis of Various Deep Learning Models for State-of-Charge Estimation in Lithium-Ion Batteries
Presented at the IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control (STPEC 2025), hosted by NIT Goa, Dec 10–13, 2025.

Certifications

Supervised Machine Learning: Regression and Classification , DeepLearning.AI Coursera Certificate	May 2025
Introduction to Power Electronics , University of Colorado Boulder Coursera Certificate	Mar 2025
Arena FIDE Master , International Chess Federation (FIDE)	Dec 2021

Technical Skills

Programming Languages: Python, MATLAB, C
Simulation and Analysis: MATLAB/Simulink, machine learning, deep learning, LSTM models, data preprocessing, model validation
Electrical Engineering: Power electronics, DC-DC converters, inverters, electrical drives, control systems, motor control, BLDC motors
Battery and Energy Systems: Li-ion batteries, battery management systems, battery energy storage systems, SoC estimation, SoH estimation
Power Devices: Wide-bandgap semiconductor devices, EV Chargers
Industrial Systems: MCCs, transformers, motors, automation systems, electrical troubleshooting, preventive maintenance

Achievements, Memberships, and Languages

- GATE 2026: All India Rank 4032
- GATE 2024: All India Rank 5151
- JEE Main 2017: All India Rank 14780
- IEEE Graduate Student Member and IEEE Power Electronics Society (PELS) Member
- Languages: English, Hindi, and Dutch